CLASS X MATHEMATICS • CHAPTER 12

Surface Areas and Volumes

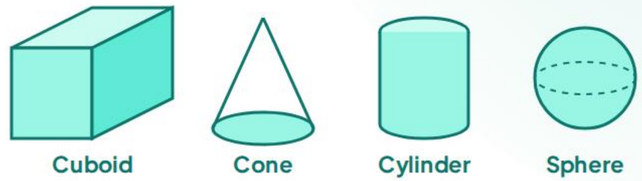
"Exhaustive point-by-point presentation covering combinations of 3D solids, deconstruction principles, visual surface boundaries, and additive volumetric analysis directly from NCERT jemh112.pdf."

Akash Srivastva (www.akashmaths.online)

Basic 3D Solids & Real-Life Combinations

The Four Fundamental Solids (Class IX Recall)

In Class IX, you explored pure geometric solids: **Cuboid**, **Right Circular Cone**, **Right Circular Cylinder**, and **Sphere / Hemisphere**.



The Real World: Combinations Everywhere!

Practical everyday structures are rarely single elementary solids:

- ✓ **Oil Tanker Truck:** A central cylinder bounded by two hemispherical caps at each end.
- ✓ **Laboratory Test Tube:** A vertical cylinder seamlessly merging into a hemispherical base.
- ✓ **Historic Monuments & Domes:** Cylindrical towers surmounted by conical spires or hemispherical domes.

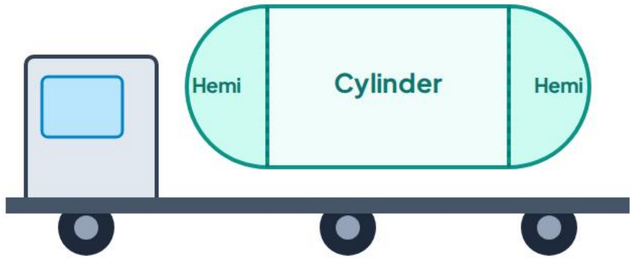


Fig 12.2: Oil Tanker Container

⚡ Quick Box • Identifying Basic Solids

Q1 Look at a common classroom pencil sharpened at one end. Which two basic three-dimensional solids make up its shape? OBSERVATION

Q2 A laboratory test tube has a hemispherical bottom. If you pull it apart conceptually, what two shapes do you get? DECONSTRUCTION

Q3 Can a standard oil tanker container be classified directly under any single elementary solid formulas studied in Class IX? Why or why not? CONCEPTUAL SHIFT

Page 162 (Textbook)

The Golden Rule of Combined Surface Areas

⚠ The Most Common Student Mistake!

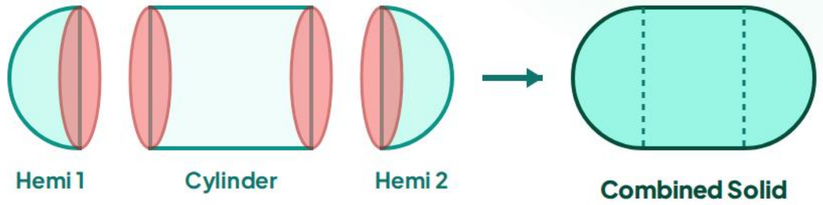
Total Surface Area (TSA) of a combined solid is NOT the sum of the TSAs of its parts! When two solids are joined flat-face to flat-face, the contacted surfaces are glued together and *cease to exist as exposed external surfaces*.

UNIVERSAL DECONSTRUCTION RULE

TSA of Combined Solid = Sum of EXPOSED Curved/Flat Areas

The "Painting Analogy" Test

Ask yourself: *"If I dip this solid in paint, which surfaces will get wet?"* Only include areas your brush can physically touch on the outside. Internal interface boundaries do NOT count!



Red Flat Circular Bases Vanish (Glued Inside)!

Visible Area = CSA(Hemi 1) + CSA(Cylinder) + CSA(Hemi 2)

⚡ Quick Box • Surface Area Deconstruction

Q1 Two identical cubes of edge 4 cm are joined end-to-end to form a cuboid. Why is the TSA of the cuboid less than 2 × (TSA of one cube)?

CORE CONCEPT

Q2 How many circular base faces disappear from the total exposed surface when two hemispheres are attached to the opposite ends of a cylinder?

INTERFACE COUNTING

Q3 If someone writes "TSA of Capsule = TSA of Cylinder + 2 × TSA of Hemisphere", what major error did they commit?

ERROR DETECTION

Pages 162, 166 (Textbook)

Cylinder with Two Hemispherical Ends

Decomposing the Dimensions

Let total length = **H**, and common diameter = **d** (radius **r = d/2**).

- ✓ Each hemispherical end has height equal to its radius: $h_{hemi} = r$.
- ✓ Both ends take up length: **r + r = 2r**.
- ✓ Length of cylindrical part: **h = H – 2r**.

TOTAL SURFACE AREA OF CAPSULE / TANKER

$$TSA = 2\pi rh + 2 \times (2\pi r^2) = 2\pi r(h + 2r)$$

Notice: 2 hemispheres combine into 1 complete sphere surface: $4\pi r^2$!

Textbook Benchmark: Exercise 12.1 Q6

Medicine Capsule: Total length = 14 mm, diameter = 5 mm (r = 2.5 mm).

- ✓ Cylinder height: h = 14 – (2.5 + 2.5) = **9 mm**.
- ✓ CSA of cylinder = $2\pi rh = 2 \times (22/7) \times 2.5 \times 9 =$ **990 / 7 mm²**.
- ✓ CSA of 2 hemispheres = $4\pi r^2 = 4 \times (22/7) \times 2.5 \times 2.5 =$ **550 / 7 mm²**.
- ✓ Total Surface Area = (990 + 550) / 7 = 1540 / 7 = **220 mm²**.

Algebraic Shortcut: $TSA = 2\pi r(h + 2r) = 2 \times (22/7) \times 2.5 \times (9 + 5) = (110/7) \times 14 =$ **220 mm²** directly!

⚡ Quick Box • Capsule & Tanker Geometry

Q1 A capsule has total length 20 mm and diameter 6 mm. What is the exact height of its cylindrical portion? MENTAL MATH

Q2 Why does the combined surface area formula of two identical hemispherical ends simplify to the surface area of a single sphere ($4\pi r^2$)? GEOMETRIC INSIGHT

Q3 If a hollow cylinder of diameter 14 cm and height 13 cm has a hollow hemisphere mounted at the bottom, how does its inner surface area formula look? HOLLOW SHELL CHECK

Cone Surmounted on a Hemisphere (Toy / Lattu)

Structural Anatomy (Fig 12.5 & 12.6)

Common radius = **r**. Entire toy height = **H**.

- ✓ Height of cone: **h = H – r**.
- ✓ Slant height of cone: **l = √(r² + h²)**.
- ✓ Flat circular face of cone joins flat circular face of hemisphere: both disappear!

TOTAL SURFACE AREA OF TOY / LATTU

TSA = CSA of Hemisphere + CSA of Cone

$= 2\pi r^2 + \pi r l = \pi r(2r + l)$

Example 1: Rasheed's Playing Top (Lattu)

Total height H = 5 cm, diameter = 3.5 cm ⇒ r = 3.5/2 = 1.75 cm.

- ✓ Cone height h = 5 – 1.75 = **3.25 cm**.
- ✓ Slant height l = √[(1.75)² + (3.25)²] = √(3.0625 + 10.5625) = √13.625 ≈ **3.7 cm**.
- ✓ CSA of hemisphere = 2 × (22/7) × 1.75 × 1.75 = **19.25 cm²**.
- ✓ CSA of cone = (22/7) × 1.75 × 3.7 = **20.35 cm²**.
- ✓ Area to colour = 19.25 + 20.35 = **39.6 cm² (approx.)**.

Colored Area ≈ 39.6 cm²

⚡ Quick Box • Cone-Hemisphere Toys

Q1 A toy has total height 15.5 cm and common base radius 3.5 cm. What is the vertical height (h) of its conical portion?

RAPID SUBTRACTION

Q2 In calculating the CSA of the conical portion, why must we calculate slant height (l) instead of directly using the vertical height (h)?

FORMULA PRECISION

Q3 If the radius of the cone is 3.5 cm and vertical height is 12 cm, calculate the slant height l in seconds (think Pythagorean triplet 7–24–25!).

MENTAL TRICK

Pages 164, 166 (Textbook)

Cube with Hemispherical Surmount / Cavity

Dual Paradox: Surmounted vs Depression

Whether a hemisphere of radius **r** is **placed on top** of a cube of edge **a** OR **scooped out** as a cavity, the total surface area formula is **IDENTICAL**!

UNIVERSAL CUBE + HEMISPHERE AREA LAW

$$\text{TSA} = 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$$

1. Total 6 faces of cube = $6a^2$
2. Remove circular contact covered/scooped = $-\pi r^2$
3. Add new exposed curved dome/bowl = $+2\pi r^2$

Example 2: Decorative Block

Cube edge a = 5 cm, hemisphere diameter d = 4.2 cm (r = 2.1 cm).

- ✓ Cube surface = 6×5^2 = **150 cm²**.
- ✓ Net added curved area = $+\pi r^2 = (22/7) \times 2.1 \times 2.1$ = **13.86 cm²**.
- ✓ Total Surface Area = 150 + 13.86 = **163.86 cm²**.

Maximum Diameter Boundary (Ex 12.1 Q4 & Q5)

The greatest diameter a hemisphere can have on a cube face without overflowing is **equal to the edge of the cube (d = a ⇒ r = a/2)**. Area = $6a^2 + \pi(a/2)^2 = \mathbf{a^2[6 + \pi/4]}$.

⚡ Quick Box • Block & Cavity Surface Areas

Q1 When a hemispherical cavity is scooped out of a wooden block, does the total surface area increase or decrease compared to the original block?

COUNTER-INTUITION

Q2 A cubical block has side 7 cm. What is the greatest possible diameter a hemisphere can have to sit entirely within one face?

GEOMETRIC LIMIT

Q3 Why does scooping out a depression add $+2\pi r^2$ rather than subtracting $2\pi r^2$ from the surface area?

VISUAL REASONING

Pages 165–166 (Textbook)

Overhanging Bases (Rocket) & Bird-Bath

Example 3: The Wooden Toy Rocket

Cone (r = 2.5 cm, h = 6 cm, l = 6.5 cm) mounted on cylinder (r' = 1.5 cm, h' = 20 cm).

Orange Region (Conical Head + Ring Base):

Area = CSA of cone + (Base of cone − Base of cylinder)

$= \pi rl + \pi(r^2 - r'^2) = 3.14 \times [16.25 + 4] = 63.585\ cm^2$

Yellow Region (Cylinder Body + Flat Bottom):

Area = CSA of cylinder + 1 base = $2\pi r'h' + \pi r'^2$

$= \pi r'(2h' + r') = 3.14 \times 1.5 \times 41.5 = 195.465\ cm^2$

Example 4: Mayank's Garden Bird-Bath

Cylinder (h = 1.45 m = 145 cm, r = 30 cm) with a hemispherical depression (r = 30 cm).

TOTAL SURFACE AREA OF BIRD-BATH

TSA = CSA(Cylinder) + CSA(Hemisphere)

$= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

$= 2 \times (22/7) \times 30 \times (145 + 30)$

$= 2 \times (22/7) \times 30 \times 175 = 2 \times 22 \times 30 \times 25$

$= 33{,}000\ cm^2 = 3.3\ m^2$

Total Area = 3.3 m²

⚡ Quick Box • Ring Bases & Depressions

Q1 In the rocket model, why can't we simply take the entire flat base of the cone as being painted orange?

OVERHANG GEOMETRY

Q2 In the bird-bath problem, why was the bottom base of the cylinder NOT included in the calculation of total surface area?

PHYSICAL CONTEXT

Q3 How do you convert 33,000 cm^2 into m^2? Why is the dividing factor 10,000 rather than 100?

UNIT CONVERSION DRILL

Page 167 (Textbook)

Volume of a Combination of Solids: Pure Additivity!

Surface Area vs Volume: The Grand Contrast

Unlike surface area where contact surfaces disappear, **volume represents internal 3-dimensional space (or capacity)**. Space never disappears upon joining!

FUNDAMENTAL VOLUME THEOREM

Total Volume = $Volume_1$ + $Volume_2$ + … + $Volume_n$

For carved/scooped cavities: Remaining Volume = Original Vol – Cavity Vol

Why Volume Calculations are Simpler

No need to worry about exposed vs hidden faces! Simply identify the individual solid shapes, compute their respective volumes using standard formulas, and sum (or subtract) them directly.

Additive Combination (Surmounted)

Toy = Hemisphere + Cone

Volume = $\frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h$

Subtractive Combination (Carved Cavity)

Block with conical depression or juice glass with dome

Remaining Volume = V(Original) – V(Cavity)

⚡ Quick Box • Additive Property of Volume

Q1 Why does the volume of two joined solids equal the direct sum of their individual volumes, while their surface areas do not?

CORE DISTINCTION

Q2 If a conical cavity is scooped out of a solid cylinder of identical base and height, what fraction of the cylinder's volume remains?

RATIO INSIGHT

Q3 A solid is formed by joining a cone of height r and radius r to a hemisphere of radius r. What is the total volume in terms of π and r?

RAPID FORMULA

Pages 167–168 (Textbook)

Industrial Shed: Cuboid + Half-Cylinder

Example 5: Shanta's Factory Shed (Fig 12.12)

Dimensions: Base = 7 m × 15 m, cuboid height = 8 m. Roof = Half-cylinder (diameter d = 7 m, radius r = 3.5 m, length = 15 m).

- ✓ **Volume of cuboid:** 15 × 7 × 8 = **840 m³**.
- ✓ **Volume of half-cylinder:** ½ × πr²h
  = ½ × (22/7) × (7/2) × (7/2) × 15 = **288.75 m³**.
- ✓ **Total Air Capacity:** 840 + 288.75 = **1128.75 m³**.

Occupational Space Deduction

When machines and personnel occupy space inside the shed:

Space occupied by machinery = **300 m³**

Space occupied by 20 workers = 20 × 0.08 m³ = **1.6 m³**

Total Occupied Volume = 300 + 1.6 = 301.6 m³

ACTUAL AIR REMAINING INSIDE SHED

1128.75 − 301.60 = 827.15 m³

⚡ Quick Box • Industrial Shed Geometry

Q1 In Shanta's shed, which dimension of the cuboid corresponds to the diameter of the half-cylindrical roof? Which corresponds to its length?

SPATIAL MAPPING

Q2 Why is the volume of the roof taken as ½ $\pi r^2 h$ rather than $\pi r^2 h$?

HALF-SOLID CHECK

Q3 If each worker occupies 0.08 m^3 and 50 workers enter the shed, by how many cubic metres is the air volume reduced?

MENTAL CALCULATION

Page 168 (Textbook)

Apparent Capacity vs Actual Capacity (Juice Glass)

Example 6: The Juice Seller's Trick Glass (Fig 12.13)

A cylindrical glass (inner diameter = 5 cm ⇒ r = 2.5 cm, height h = 10 cm) has a **raised hemispherical bottom** inside it, cheating the customer out of liquid volume!

Apparent Capacity

What the customer sees (pure cylinder):

$V = \pi r^2 h$

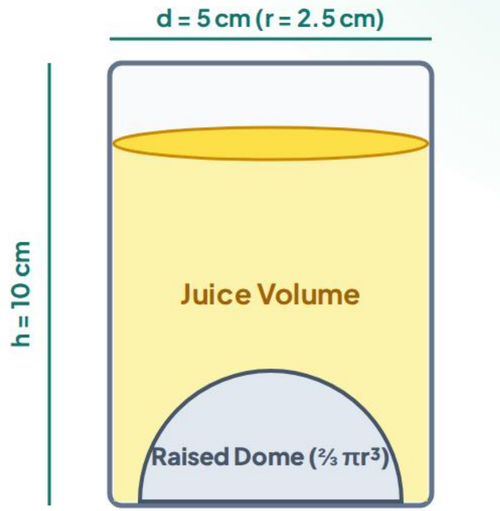
$= 3.14 \times 2.5^2 \times 10 =$ **196.25 cm³**

Actual Capacity

What the glass actually holds:

$V = \pi r^2 h - \frac{2}{3}\pi r^3$

$= 196.25 - 32.71 =$ **163.54 cm³**



Actual Capacity = Apparent − Dome Vol

⚡ Quick Box • Apparent vs Actual Capacity

Q1 What is the formula difference between the apparent capacity and actual capacity of the juice glass? FORMULA RECALL

Q2 By what volume (in cm^3) was the customer cheated if the raised dome has radius 2.5 cm and takes $\pi = 3.14$? DEFICIT VALUE

Q3 If the raised portion were conical instead of hemispherical with the same height r = 2.5 cm, would the customer lose more juice or less? COMPARATIVE VOLUME

Toy Circumscribed by a Right Circular Cylinder

Example 7: Structure & Clearances (Fig 12.14)

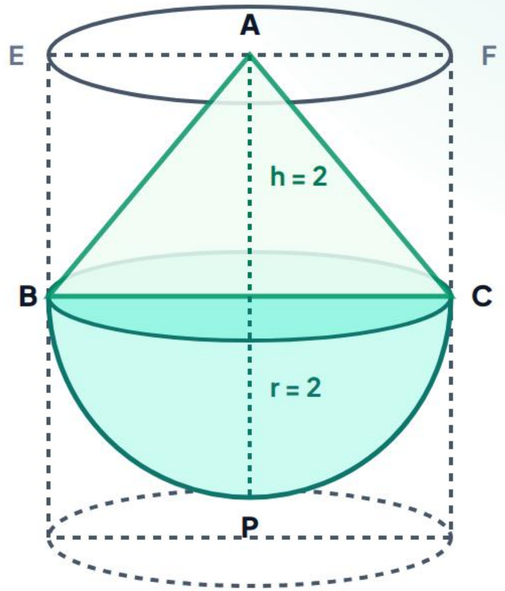
A solid toy is a hemisphere (r = 2 cm) surmounted by a cone (r = 2 cm, h = 2 cm). A right circular cylinder circumscribes this entire toy snugly.

Critical Dimension Linking

- ✓ Radius of circumscribing cylinder = radius of toy = **2 cm**.
- ✓ Height of cylinder = Cone height + Hemisphere radius = 2 + 2 = **4 cm**.
- ✓ **Volume of toy:** $\frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi r^2(2r + h) = \frac{1}{3} \times 3.14 \times 4 \times 6 =$ **25.12 cm³**.

VOLUME CLEARANCE DIFFERENCE

V(Cylinder) − V(Toy) = 50.24 − 25.12 = 25.12 cm³



⚡ Quick Box • Circumscribing Cylinders

Q1 When a cylinder snugly circumscribes a cone on a hemisphere, what two lengths add together to equal the cylinder's total height?

HEIGHT LINKING

Q2 In Example 7, why did the volume of the cylinder turn out to be exactly double the volume of the toy? (50.24 vs 25.12 cm^3)

MATHEMATICAL RATIO

Q3 If a solid cylinder full of water contains a submerged solid, how is the volume of water left related to the volume of the submerged object?

DISPLACEMENT PRINCIPLE

EXERCISES 12.1 & 12.2 • PRACTICAL MODELS

Pages 166–170 (Textbook)

Rich Contextual Engineering Models in NCERT

Wooden Pen Stand

Cuboid (15 × 10 × 3.5 cm) with **4 conical depressions** for pens (r = 0.5 cm, depth h = 1.4 cm).

$V = V(\text{Cuboid}) - 4 \times V(\text{Cone})$

Gulab Jamun & Syrup

Cylinder + 2 hemispherical ends (l = 5 cm, d = 2.8 cm). Syrup = **30% of total volume** for 45 pieces!

Percentage Multiplier

Solid Iron Pole Mass

Cylinder (h = 220 cm, d = 24 cm) surmounted by smaller cylinder (h = 60 cm, r = 8 cm). Density = **8 g / cm³**.

Mass = Total Volume × Density

Submerged Lead Shots

Inverted cone filled with water. Spherical lead shots dropped: **¼ of water overflows**. Find number of shots n.

$n \times V(\text{Sphere}) = \frac{1}{4} V(\text{Cone})$

Teacher's Strategy: Every question connects geometry with physics (density, mass), commerce (cost per m²), or displacement (Archimedes' principle).

⚡ Quick Box • Contextual Problem Drills

Q1 When lead spheres are dropped into a full vessel of water, how do we relate the volume of displaced water to the volume of the spheres?

ARCHIMEDES PRINCIPLE

Q2 In calculating the mass of the composite iron pole, why must we compute the entire volume in cm^3 before multiplying by 8 g/cm^3?

MASS-VOLUME RELATIONSHIP

Q3 In the pen stand problem, why do we subtract the volume of the conical depressions, but would ADD their curved surface area if finding the surface area to be polished?

DEEP CONTRAST CHECK

EXERCISE HIGHLIGHTS • PART 1

Page 166 (Textbook)

Worked Highlights: Hollow Vessel & Solid with Cavity

EXERCISE 12.1 • QUESTION 2

Inner Surface Area of Hollow Vessel

Given: Hollow cylinder mounted on hollow hemisphere. Diameter = 14 cm (r = 7 cm), total height = 13 cm.

- ✓ Height of cylinder: h = 13 – 7 = **6 cm**.
- ✓ Inner surface area = CSA of cylinder + CSA of hemisphere
- ✓ $= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$
- ✓ $= 2 \times (22/7) \times 7 \times (6 + 7) = 44 \times 13 =$ **572 cm²**.

Inner Area = 572 cm²

EXERCISE 12.1 • QUESTION 8

Cylinder with Hollowed-out Conical Cavity

Given: Solid cylinder (h = 2.4 cm, d = 1.4 cm ⇒ r = 0.7 cm) with conical cavity of same dimensions.

- ✓ Slant height $l = \sqrt{(0.7^2 + 2.4^2)} = \sqrt{(0.49 + 5.76)} = \sqrt{6.25} =$ **2.5 cm**.
- ✓ TSA = CSA(Cyl) + 1 base area + CSA(Cone)
- ✓ $= 2\pi rh + \pi r^2 + \pi rl = \pi r(2h + r + l)$
- ✓ $= (22/7) \times 0.7 \times (4.8 + 0.7 + 2.5) = 2.2 \times 8.0 =$ **17.6 cm² ≈ 18 cm²**.

Nearest TSA ≈ 18 cm²

EXERCISE HIGHLIGHTS • PART 2

Page 170 (Textbook)

Worked Highlights: Solid Cone in Terms of π & Pen Stand

EXERCISE 12.2 • QUESTION 1

Volume of Solid in Terms of π

Given: Cone standing on hemisphere. Common radius r = 1 cm, cone height h = r = 1 cm.

- ✓ Volume of solid = Volume of cone + Volume of hemisphere
- ✓ $= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = \frac{1}{3}\pi(1)^2(1) + \frac{2}{3}\pi(1)^3$
- ✓ $= \frac{1}{3}\pi + \frac{2}{3}\pi =$ **π cm³**.

Exact Volume = π cm³

EXERCISE 12.2 • QUESTION 4

Wooden Pen Stand with 4 Conical Depressions

Given: Cuboid (15 × 10 × 3.5 cm), 4 conical depressions (r = 0.5 cm, h = 1.4 cm).

- ✓ Volume of cuboid = 15 × 10 × 3.5 = **525 cm³**.
- ✓ Volume of 4 depressions = 4 × [⅓ × (22/7) × 0.5 × 0.5 × 1.4] = **1.47 cm³**.
- ✓ Volume of wood = 525 − 1.47 = **523.53 cm³**.

Wood Volume = 523.53 cm³

Page 170 (Textbook)

Chapter 12 Master Formula & Concept Matrix

Surface Area Guidelines

- ✓ **Fundamental Rule:** Surface Area is purely EXPOSED area. Glued contact boundaries disappear!
- ✓ **Cylinder + 2 Hemispheres:**
  TSA = $2\pi rh + 4\pi r^2 = 2\pi r(h + 2r)$
- ✓ **Cone + Hemisphere:**
  TSA = $\pi rl + 2\pi r^2 = \pi r(l + 2r)$ [where $l = \sqrt{(r^2 + h^2)}$]
- ✓ **Cube ± Hemisphere (Surmount / Depression):**
  TSA = $6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$

Volume & Capacity Guidelines

- ✓ **Fundamental Rule:** Volume is strictly ADDITIVE. Space never vanishes upon joining!
- ✓ **Attached Composite Volume:**
  $V_Total = V_1 + V_2 + \ldots + V_n$
- ✓ **Carved/Scooped Cavity Volume:**
  V_Remaining = V(Outer Solid) − V(Cavity)
- ✓ **Liquid Displacement:**
  Volume of Overflowing Liquid = Volume of Submerged Body

NCERT CLASS X MATHEMATICS

Thank You!

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